Chapter 1

The Principles and Practice of Economics

ECON 003
UC Riverside · Summer 2026



Where This Course Is Going

TODAY

Ch. 1 — the foundation

Weeks 1–2

Optimization, supply & demand

Weeks 3–4

Firms, competition, trade

Week 5

Policy, taxes, monopoly

Everything in this course rests on three ideas — and you will meet all three in the next 70 minutes.

Optimization. Equilibrium. Empiricism.

Every model we build for the rest of the term is some version of these three. Chapter 1 is the shortest chapter in the book, and the one that everything else is built on top of.

Today's Roadmap

01

The Scope of Economics

Choice, scarcity, and what economists actually study

02

The First Principle: Optimization

Trade-offs, budget constraints, opportunity cost

03

The Second Principle: Equilibrium

When nobody wants to change what they are doing

04

The Third Principle: Empiricism

Using data to find out what is actually true

PART 1

The Scope of Economics

Economics Is About Choice, Not Money

WHAT PEOPLE ASSUME

Economics is about money — prices, stock markets, interest rates, the Federal Reserve.

A subject about finance, for people who want to work in finance.

WHAT IT ACTUALLY IS

Economists study human behavior.

Choice — not money — is the unifying feature of everything economists study.

Price is one very important variable that helps explain behavior. But it is the behavior we are after: the decision to lease a car, to skip class, not to wear a seat belt, or how fast to take a hairpin corner.

Who Is an Economic Agent?

An economic agent is any individual or group that makes choices.

INDIVIDUALS

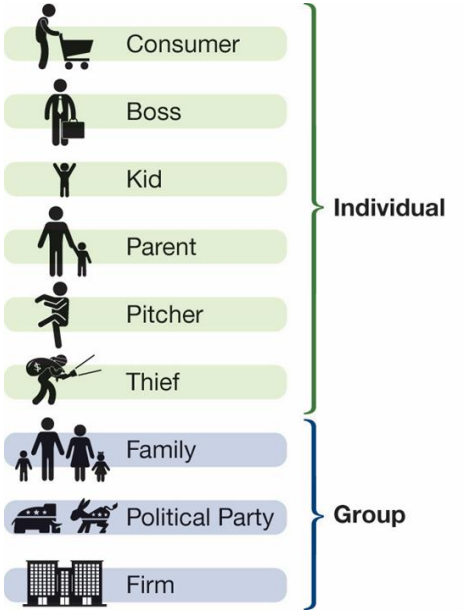
a consumer · a student · a worker
a voter · a parent · a criminal
a senator · a business leader

GROUPS

a government · an army · a firm
a university · a labor union
a family · a sports team

And you. You are an economic agent, making an enormous number of choices every day.

Agents and Their Choices



Economic agent	One choice they make
A consumer	Which cell phone to purchase
A parent	Whether and how to reward a teenager for good behavior
A student	To attend class or to skip it
A citizen	Whether to vote — and which candidate to support
A worker	To do the job or to slack off
A business leader	Where in the world to open a new production facility
A senator	To vote for or against a Supreme Court nominee

Every row is a scarce resource being allocated by somebody making a choice.

Scarcity: The Problem Underneath Everything

A bowl of chocolates is passed around a class of 100 students.

There are exactly 100 chocolates. Everyone is told: take as many as you want.



SCARCE RESOURCES VS SCARCITY

SCARCE RESOURCES

Things that people want, where the quantity people want exceeds the quantity available.

A resource does not have to be luxurious: toilet paper, subway seats, clean drinking water, anything that is limited.

SCARCITY

The situation of having unlimited wants in a world of limited resources.

The world does not have enough resources to give everyone everything they want, for free.

Scarcity is why choice is unavoidable — and choice is why economics exists.

The Definition of Economics

Economics studies how agents make choices among scarce resources and how those choices affect society.

Positive vs. Normative Economics

POSITIVE ECONOMICS

Describes what people actually do.

“Some students took more than one, so not everyone got a chocolate.”

Descriptive. Testable against data.

NORMATIVE ECONOMICS

Recommends what people — or society — ought to do.

“Each student should take just one so that everyone gets a chocolate.”

Advisory. Requires a value judgment.

A normative claim always hides a decision about what counts as fair — and you have to defend it.

Microeconomics vs. Macroeconomics

MICROECONOMICS

The study of how individuals, households, firms, and governments make choices — and how those choices affect prices, the allocation of resources, and the well-being of other agents.

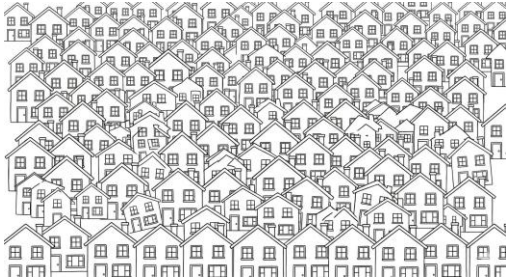
One piece of the economy: the market for coal-fired electricity.



MACROECONOMICS

The study of the economy as a whole.

Economy-wide phenomena:
· the growth rate of total output
· the inflation rate
· the unemployment rate



The Three Principles of Economics

What separates economics from the other social sciences.

OPTIMIZATION

Making the best choice you can with the information you have.

People do not always succeed — but they try.

EQUILIBRIUM

Everyone is optimizing at once, so nobody would benefit by changing their own behavior.

EMPIRICISM

Using data to work out whether the theory actually matches how people behave.

Anthropology, history, political science, psychology and sociology all study human behavior too. These three commitments are what make the economist’s version of that project distinctive.

PART 2

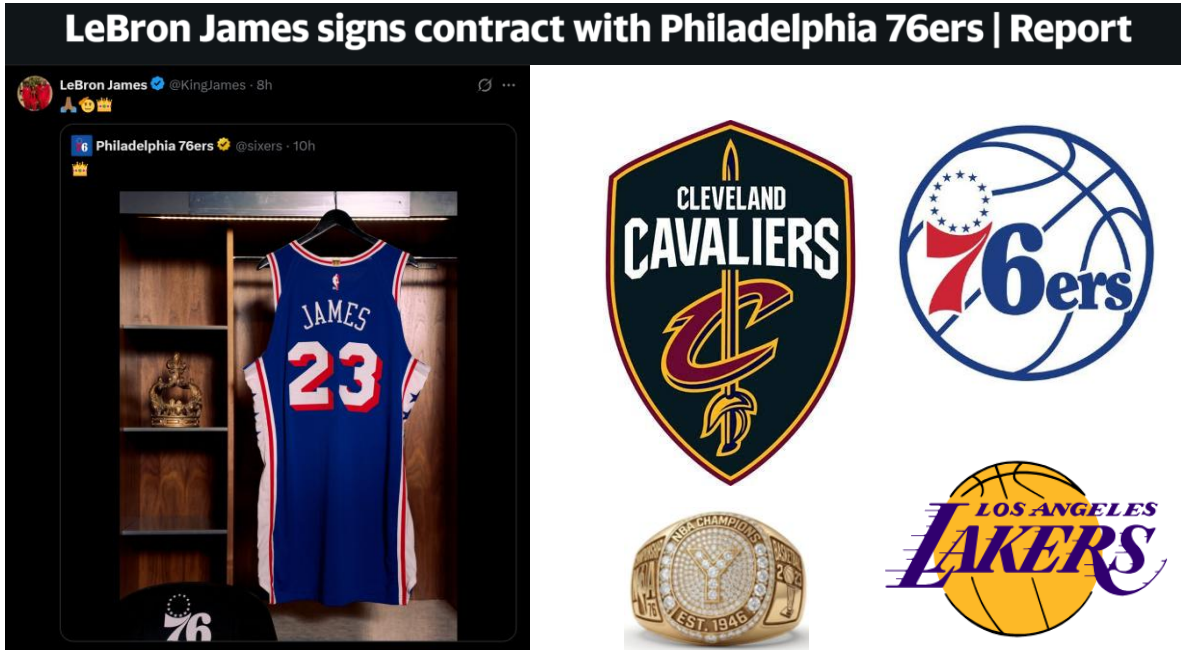
The First Principle: Optimization

What Optimization Means

Optimization = making the best feasible choice with the information available.

Every Optimization Involves a Trade-off

Trade-offs arise when some benefits must be given up in order to gain others.



Opportunity Cost

Opportunity cost is the value of the best alternative use of a resource.

WHAT LEBRON JAMES CHOSE · July 2026

Philadelphia 76ers — $8M in two years

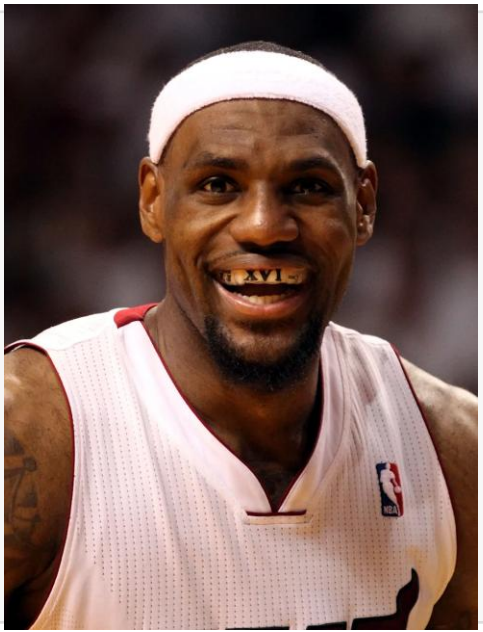
Two years, $8M total, veteran minimum, player option



HIS NEXT-BEST OPTION

Golden State Warriors — about $75M

Opportunity cost ≈ $67 million



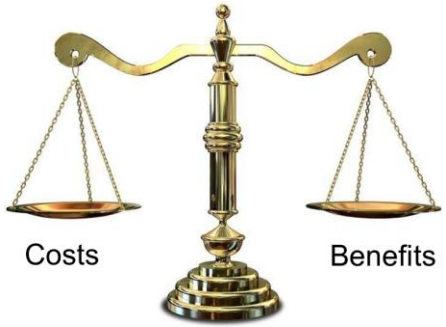
Cost-Benefit Analysis and Trade-offs

A calculation that identifies the best option by summing benefits and subtracting costs.

Net benefit = total benefits − total costs

COSTS OF GOING TO 76ERS

$67 million.



BENEFITS OF GOING TO 76ERS

Probably a great chance to win the champion a fifth time.

Is Tiktok Free?

You pay $0 to open the app.

What do you give up when you spend an hour on TikTok?
If you put a value on the activities you gave up, which is the highest?

PAID WORK

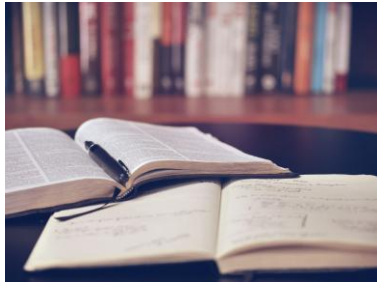
An hour of wages you did not earn



$17

STUDY

An hour you did not spend on this course



$20

OTHER RECREATION

Video games, soccer, sleep, a friend, a walk



$25

Putting a Price on Your Attention

97 min

Per day on Tiktok,
average Tiktok user, 2025

$16.90

Opportunity cost
of an hour of time in California

$9,970

Per U.S. adult,
per year

≈ 1.5 hour / day × 365 days × $17 / hour ≈ $10,000 per year

Across all U.S. adults, the opportunity cost of time on social media came to roughly $1.2 trillion in 2019.

So, no — Tiktok is not free. It is one of the most expensive things many people consume.

Budget Constraints

A budget constraint is the set of things a person can choose to do (or buy) without breaking her budget.

MONEY

$60 to spend on food this week.

TIME

24 hours in a day.
168 in a week.

ATTENTION

You cannot read two things at once, however much you want to.

Planning a $10,000 Trip to Europe

Same $10,000, same 14 days. Plan A sees five cities; Plan B sees two.



Item	Plan A — breadth (5 cities)	Plan B — depth (2 cities)
Flights	$1,300	$1,300
Trains between cities	$650	$150
Hotels	$2,800	$4,000
Food & drink	$1,250	$1,250
Sights, tours, museums	$3,000	$1,000
Shopping & extras	$1,000	$2,300
TOTAL · 14 days	**$10,000**	**$10,000**



BUDGET CONSTRAINT

Both columns cost exactly $10,000 over the same 14 days.

TRADE-OFF

Breadth pays for trains and tickets. Depth pays for hotels and shopping.

OPTIMIZING

Comfort vs Experience



PART 3

The Second Principle: Equilibrium

What Is Equilibrium?

Equilibrium is a situation in which nobody would benefit by changing his or her own behavior.

THE LINK TO OPTIMIZATION

Equilibrium is what you get when
everyone is optimizing at the same time.
One person optimizing is not enough.

Equilibrium does not mean “good.”
It means nothing is pushing anyone to change.



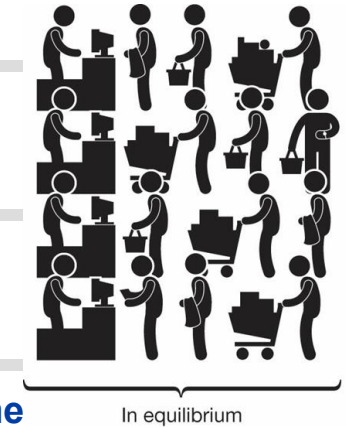
Checkout Line

01 A line is shorter than the others

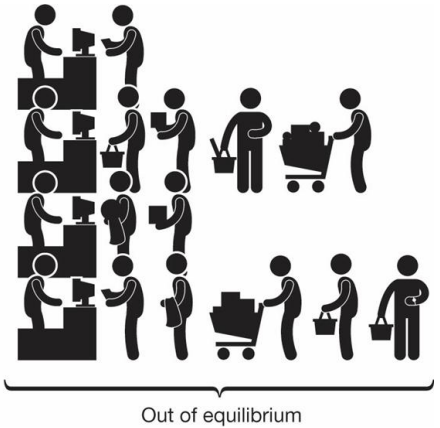
02 A line is longer than the others

03 Short lines fill up; long lines drain

04 Every line takes about the same time



In equilibrium



Out of equilibrium

The Free Rider Problem

A free rider enjoys the benefits of a situation without incurring the costs.



IS HOMER IN EQUILIBRIUM?

Yes. No incentive to change.



IS PBS IN EQUILIBRIUM?

No. They have to force an incentive onto the viewers.

Markets have no mechanism to decide what is **fair**.

PART 4

The Third Principle: Empiricism

Empiricism: Testing Ideas with Data

Empiricism is evidence-based analysis — using data to test whether our theories about human behavior match what people actually do.

Where should McDonald's put its next restaurant?







Predict, then measure. The gap between them is where economics actually happens.

Correlation Is Not Causation

“We often see fast-food chains together. We should locate our next McDonald’s right next to a Burger King.”



Photo: Chris Light · CC BY-SA 4.0 · via Wikimedia Commons (curid 95567816)

The correlation is real but not the cause.

Locations near a Burger King really do post higher sales.

The causation is otherwise.

Traffic causes good sales.

Highway exits, traffic counts, busy corridors have high customer volume.

They can share supply chains as well.

Chapter 1: The Full Picture

	The core claim	Key terms	Story
Scope	Wants exceed resources, so choice is unavoidable	Economic agent, scarce resources, scarcity, positive vs. normative, micro vs. macro	The bowl of chocolates
Optimization	People try to pick the best feasible option	Trade-off, budget constraint, opportunity cost, cost-benefit analysis, net benefit	Lebron James' choice for next team.
Equilibrium	Everyone optimizes at once; nobody gains by changing alone	Equilibrium, free rider problem	Homer watching PBS.
Empiricism	Data decides whether the theory is right	Empiricism, causation vs. correlation	Choosing a location for McDonald's

Four rows. Everything else this term is one of them, in more detail.

KEY TAKEAWAYS

Chapter 1: Principles and Practice

- Economics is the study of choice under scarcity — not the study of money
- Positive economics describes; normative economics prescribes, with a value judgment
- Optimization: pick the best feasible option, and measure cost as opportunity cost
- Equilibrium: everyone optimizing at once, so nobody gains by changing alone
- Empiricism: data decides — and correlation is not causation